COMPLETE PROJECT HANDOFF

Rojin AI Can-Lid Inspection | Conversation history, verified technical state, and proposal continuity

Artifact / status	Verified detail
Primary model	YOLO11n-cls: yolo11n_cls_masked_random80_v2
Current deliverable	English industrial demo proposal v1 (PDF + editable Word)
Project root	D:\rojin_lid_ai (Windows; user machine)
Dataset origin	Public/external TinplateLidDATA2024; NOT yet validated on Rojin factory images
Data coverage	1,307 segmented images; 943 train / 103 validation / 261 untouched test
Test outcome	110 GOOD / 148 DEFECTIVE / 3 REVIEW; 257/261 correct decisive actions
Throughput	4.185 s for 261 pre-masked images = 62.36 images/s (measured test-script scope)
Evidence basis	Project ZIP, exported metrics/CSV, training console log, integrity audit, reference PDFs, and conversation

Prepared: 2026-10-10

What this document is: A complete practical handoff summary of the accessible project conversation and verified artifacts. It is not a word-for-word export of every historic chat message; some earlier exchanges were available only as summaries. Facts from files are separated from suggestions and unverified plans.

راهنمای کوتاه برای شروع چت جدید

این فایل، خلاصهٔ جامع و مستند پروژه «تشخیص سالم یا معیوب بودن درب قوطی رب روژین» است. اگر در یک گفت‌وگوی جدید آن را بارگذاری کردی، از دستیار بخواه ابتدا کل فایل را بخواند و فقط براساس وضعیت واقعی پروژه ادامه دهد.

مدل نهایی پروژه yolo11n_cls_masked_random80_v2 است. این مدل یک دموی آفلاین برای درب قوطی است، نه سیستم کامل بازرسی کارخانه. دیتاست ۱۳۰۷ تصویری عمومی و خارجی است و هنوز صحت عملکرد مدل روی تصاویر واقعی خط تولید روژین اثبات نشده است.

تقسیم داده‌ها: ۹۴۳ تصویر آموزش، ۱۰۳ تصویر اعتبارسنجی و ۲۶۱ تصویر تست مستقل. همه تصاویر مرجع حفظ شده‌اند و طبق بررسی SHA-256 هیچ فایل عیناً تکراری بین آموزش و تست وجود ندارد؛ بااین‌حال شباهت ظاهری یا فریم‌های متوالی هنوز به‌طور مستقل بررسی نشده‌اند.

نتایج دمو: دقت تصمیم کلی ۹۸٫۴۷ درصد در تست فعلی؛ ۱۱۰ تصویر GOOD، ۱۴۸ تصویر DEFECTIVE و ۳ تصویر REVIEW. سرعت ثبت‌شده ۶۲٫۳۶ تصویر بر ثانیه مربوط به تصاویر ازقبل ماسک‌شده است و شامل تصویربرداری و جداسازی کامل درب از تصویر خام نیست.

پروپوزال انگلیسی نسخه اول قبلاً در دو قالب PDF و Word ساخته شده است. پیش از نهایی‌سازی برای مدیریت روژین، متن، تصاویر و معیارهای عملکرد باید با دقت بازبینی شوند. قابلیت‌های کنترل لیبل، بدنه و سایر محصولات صرفاً برنامه توسعه آینده پس از پذیرش دمو هستند.

فایل‌های مهم همراه این PDF: rojin_proposal_review_package.zip و Rojin_AI_Can_Lid_Inspection_Proposal_v1.pdf (در صورت دسترسی). ZIP اصلی داده‌ها برای فهم پروژه در گفتگوهای بعدی ضروری نیست.

متن پیشنهادی برای ارسال در چت جدید

> “Please read the attached Rojin Lid AI Complete Project Handoff PDF in full, treat the verified model/data/test/proposal facts as the baseline, distinguish the public-dataset laboratory demo from real Rojin factory validation, and help me continue the English industrial proposal. Ask only for information that is truly missing; do not reset the project or confuse the old 60-epoch model with yolo11n_cls_masked_random80_v2.”

01 Executive state: what is established

Project goal. Develop a Windows/offline computer-vision proof of concept for assessing tomato-paste tinplate can lids, producing two class probabilities and one operational action: GOOD, DEFECTIVE, or REVIEW. Later expansion to can body, label, printing and other Rojin products is a proposal, not deployed capability.

Ground-truth origin. TinplateLidDATA2024 is a PUBLIC/EXTERNAL dataset that resembles the intended Rojin tomato-paste lids. No assertion that the 1,307 images were captured inside Rojin. This must be transparent in the English proposal.

Item	Current verified state
System	Windows; Python virtual env; Ultralytics YOLO11n-cls; CUDA GPU
GPU / CPU / RAM	NVIDIA GeForce RTX 5060 (reported), Intel i5-12400, 32 GB RAM
Dataset	1,307 unique masked image files, with binary class labels
Final weights	D:\rojin_lid_ai\results\yolo11n_cls_masked_random80_v2\weights\best.pt
Final training	61 epochs completed; early stop; best validation at epoch 41
Final test	261 pre-masked held-out images; 98.47% all-image operational accuracy
Preview output	results\random20_final_test\{good,defective,review}
Proposal	Rojin_AI_Can_Lid_Inspection_Proposal_v1.pdf/.docx, 12-page English draft

Never misstate these distinctions

- YOLO is trained on TWO labels: good and defective. REVIEW is created by post-processing thresholds, not a third learned class.
- The reported 98.47% includes REVIEW decisions in the denominator as not correct. This is not necessarily raw classifier top-1 accuracy.
- The measured 62.36 images/s does not prove 62.36 finished products/s on a live plant conveyor. Full acquisition + segmentation + controls were not timed.
- No byte-identical train/test overlap was found; perceptual near-duplicate or same-session frame leakage is not ruled out.
- The final 80/20 model cannot be replaced with the earlier masked_v1 best.pt for this reshuffled holdout, due to possible test leakage.

02 User priorities and communication rules

- Conversation is primarily in Persian; proposal to Rojin must be in professional English.
- Emphasize accurate, executable Windows/PowerShell instructions with full paths and robust complete scripts. Previous deletion and multiprocessing mistakes increased concern about code safety.
- Preserve an untouched 20% independent test set; use the 80% pool exclusively for training and validation. Do not tune thresholds on test.
- Images are real evidence. Do not use stock or other vendors' device photos as proof that the user built or deployed those machines.
- Primary inspection risk is FALSE ACCEPT: defective can lid approved as GOOD. Also report false rejects, review coverage, throughput, and confidence scores.
- Target at least 25 end-to-end images/s; do not conflate GPU inference with actual pipeline throughput.
- Demo scope must remain lid inspection; body, label, packaging, other products, camera/PLC integration belong to optional future phases.

- Keep the internal SHA-256 audit out of the customer-facing proposal unless a short methodology mention is helpful; it was commissioned mainly for internal assurance.

03 Project filesystem on the user's computer

```
D:\rojin_lid_ai\
  .venv\                    # local virtual environment (exclude from archive)
  src\                      # pipeline, train, test, audit, packaging scripts
  data\raw\TinplateLidDATA2024\
  data\processed\TinplateLidDATA2024_masked_v1\
  data\processed\TinplateLidDATA2024_masked_random80_20\
  data\processed\TinplateLidDATA2024_masked_random80\
    train\{good,defective}    # 943 train images
    val\{good,defective}      # 103 validation images
  data\processed\TinplateLidDATA2024_masked_random20\
    test\{good,defective}     # 261 holdout images
  results\yolo11n_cls_masked_random80_v2\
    weights\best.pt
    weights\last.pt           # observed in training console
    args.yaml, results.csv, results.png, confusion_matrix.png
  results\random20_final_test\
    good\, defective\, review\
    predictions.csv, metrics.json, classification_report.txt,
    confusion_matrix.png
  results\dataset_integrity_audit\
    integrity_summary.json, issues.txt, file_inventory.csv
  models\, weights\, yolo11n-cls.pt
```

Important: Ultralytics .train(data=...) must point to the PARENT directory containing train and val. It must not point directly to random80/train.

04 Dataset origin, masking, and preparation

Raw data and classes

TinplateLidDATA2024 originally contains 1,307 examples: 571 good and 736 defective. Its labels come from directory names, not segmentation polygons. No YOLO-seg annotations were needed to train the classification model.

Masking / ROI extraction

src/01_build_masked.py --full processed all 1,307 images and wrote masked lid crops with a black background to data/processed/TinplateLidDATA2024_masked_v1, plus masks, overlays, audit sheets and reports. The saved summary.json reports expected_images=1307, ok=1307, review=0 and all_splits_separate=true.

Caveat: “OK” is only the script’s automatic quality flag. The user visually noticed approximately 30 imperfect masks in the earlier audit. A failure-free mask generation count is not proof that every boundary and subtle edge defect is preserved. Overlay/audit images must not be used as training data.

Original split versus final split

Version	Good	Defective	Total	Use
Original earlier train	457	588	1045	Historical only
Original earlier val	85	112	197	Historical only
Original earlier test	29	36	65	Historical only
NEW 80% train pool (before val)	457	589	1046	Source for current train+val
NEW 20% test holdout	114	147	261	FINAL independent test
Current train	412	531	943	Final model training
Current val	45	58	103	Final model early stopping

The new stratified image-level split was created by src/09_create_final_dataset_split.py using seed 42: 80% (1046 images) and untouched 20% (261 images). src/10_create_val_from_random80.py extracted a stratified 10% validation portion from the 80% pool; it used 45 good and 58 defective val images and left 943 actual train images. Splits were derived from the masked_v1 dataset.

Source reports

- data/processed/TinplateLidDATA2024_masked_v1/summary.json
- .../TinplateLidDATA2024_masked_random80_20/split_report.json
- .../TinplateLidDATA2024_masked_random80/val_report.json
- .../TinplateLidDATA2024_masked_random80_20/split_manifest.csv

05 Dataset separation audit: verified and limited

The user ran src/11_verify_dataset_integrity.py (read-only SHA-256 audit) on 2026-10-09. It compared the parent masked_random80_20 reference against the actual random80 train/val and random20 test. The uploaded integrity_summary.json and issues.txt agree: RESULT PASS.

Audit criterion	Observed
All source images covered	TRUE (1307/1307)
Original 80/20 membership preserved	TRUE
Train vs test identical content hashes	0
Train vs validation identical content hashes	0

Audit criterion	Observed
Duplicate content in active dataset	0
Duplicate filenames in active dataset	0
Missing images vs reference	0
Extra images vs reference	0
Overall audit result	passed = true

Limitation. SHA-256 catches byte-identical files, not visually similar near-duplicates. Names encode timestamps and capture rates (e.g. 21Hz/23Hz/60Hz/709RPM); nearby frames or shared production/capture sessions could have perceptual overlap. A future stronger evaluation should consider perceptual hashing and grouped-by-session/batch splits or genuinely new Rojin production imagery. Do not present the random image-level test as independent proof of real-world factory generalization.

06 Training: definitive yolo11n_cls_masked_random80_v2

The first successful older masked_v1 run was not the final model. A distinct final training run started from pretrained yolo11n-cls.pt with the random80 train/val folders. GPU acceleration was confirmed; Windows multiprocessing was stabilized with workers=0.

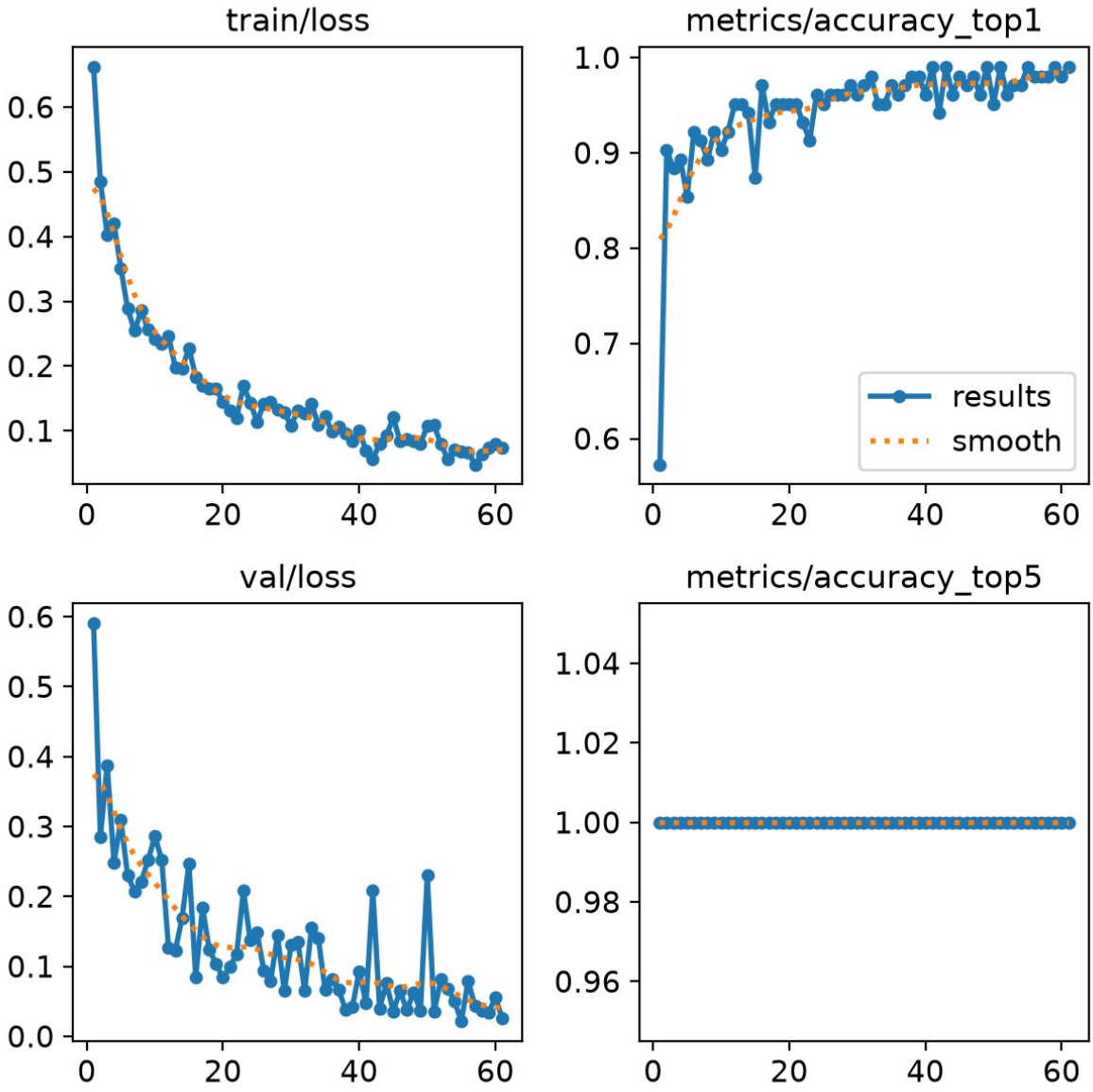
Key hyperparameter / outcome	Actual verified value
Model	YOLO11n-cls (pretrained yolo11n-cls.pt)
Task	classify; 2 classes good / defective
Dataset	data/processed/TinplateLidDATA2024_masked_random80
Images / resolution	943 train; 103 val; imgsz=640
Batch / workers	32 / 0
Epoch cap / patience	100 / 20
Optimizer / LR	AdamW / lr0=0.001
Device / automatic precision	CUDA device=0 (RTX 5060) / amp=True
Train seed / deterministic	seed=0, deterministic=True (NOT split seed=42)
Epoch outcome	61 epochs, 0.542 hours; early stop
Best epoch / validation top-1	41 / about 0.99 (103 validation samples)
Best weights	results/yolo11n_cls_masked_random80_v2/weights/best.pt

Source: src/02_train_masked.py, results/yolo11n_cls_masked_random80_v2/args.yaml and the full console log in Pasted text(7).txt. Real arguments were captured in args.yaml. Do not claim a full 100 epochs ran; early stopping at epoch 61 is normal, not a failure.

Actual training augmentations deserve attention

The saved args.yaml includes auto_augment=randaugment, erasing=0.4, scale=0.5, fliplr=0.5, translate=0.1. These are Ultralytics classification defaults/active settings; they may crop or obscure tiny lid-edge defects. This does NOT invalidate the logged experiment, but should be re-examined for a safety-critical product and future training. The end-to-end proposal should not falsely claim edges are always preserved in every augmentation.

Training plot from final model



Source: results/yolo11n_cls_masked_random80_v2/results.png in packaged project.

07 Inference and the three-action decision policy

The FINAL model provides normalized softmax-like classification scores between 0 and 1: p_good and p_defect. A separate downstream decision rule classifies the result as an operational action. These thresholds are NOT training parameters and are NOT learned as a third class.

```
GOOD_THRESHOLD = 0.20
REVIEW_THRESHOLD = 0.50

def decide(p_defect):
  if p_defect < GOOD_THRESHOLD:
    return "good"
  elif p_defect <= REVIEW_THRESHOLD:
    return "review"
  else:
    return "defective"
```

Condition	Final action
p_defect < 0.20	GOOD
0.20 <= p_defect <= 0.50	REVIEW
p_defect > 0.50	DEFECTIVE

Current test script src/03_final_test_random20.py uses MODEL_PATH=.../random80_v2/weights/best.pt and TEST_PATH=.../masked_random20/test. It copies each original MASKED test image (without confidence text overlay) into results/random20_final_test/{good,defective,review} and writes a per-image predictions.csv with image, true_label, p_good, p_defect and decision.

Threshold contrast. Older scripts src/03_evaluate_masked.py, evaluate_demo25.py, evaluate_internet25.py and test_external.py used LOW=0.15 and HIGH=0.45 (with <= and >=) for three-action output. Some legacy binary metrics used p_defect >= 0.5. src/08_calibrate_validation.py explores threshold choices on masked_v1 VALIDATION, not the latest split. Never confuse these rules or claim 0.20/0.50 was optimized for random80_v2 unless proven by a validation-only calibration experiment.

08 Final held-out test: precise verified results

The test uses the 261 held-out masked images: 114 genuinely good and 147 genuinely defective. Prediction batch size = 32 on CUDA device=0, half=True (deprecated warning in current Ultralytics version). It processed all 261; there was no fatal error in the final run.

Ground truth \ system decision	GOOD	DEFECTIVE	REVIEW	Row total
Actually good	110	1	3	114
Actually defective	0	147	0	147
Column total	110	148	3	261

Operational metrics and correct denominators

Metric	Value	Interpretation
Total evaluated	261	Held-out public images
All-image correct-decision accuracy	257/261 = 98.47%	REVIEW not scored as a correct class
Decisive coverage	258/261 = 98.85%	All non-REVIEW decisions
Decisive-only accuracy	257/258 = 99.61%	Exclude 3 REVIEW images from denominator
Observed false accepts	0	Actually defective -> GOOD
Observed false rejects	1	Actually good -> DEFECTIVE
Defect recall (immediate DEFECTIVE decisions)	147/147 = 100%	In this limited test only
Defect precision (among DEFECTIVE decisions)	147/148 = 99.32%	One false reject in DEFECTIVE group
Review count / rate	3 / 261 = 1.15%	All 3 REVIEW images are actually good
Measured time / throughput	4.185 s / 62.36 images/s	Specific pre-masked batch-processing script

Important: The older exported metrics.json also contains precision_macro=0.6644, recall_macro=0.655, f1_macro=0.6596. Those macro values are misleading for the industrial two-label ground truth because sklearn treated “review” as a third class with zero actual support. Keep REVIEW as an abstention / human-routing outcome and report binary or selective metrics with clear denominators.

Why 98.47% is not the complete AI model accuracy

All-image system decision accuracy is (110 good correctly accepted + 147 defects correctly rejected)/261 = 257/261. It penalizes all three REVIEW results as not automatically correct; one additional error is a good lid mis-rejected. To estimate raw YOLO binary top-1 accuracy, calculate from p_good/p_defect argmax explicitly on a saved untouched test set, rather than re-labeling REVIEW arbitrarily.

Representative cases pulled from the original CSV

True	p_defect	Decision	Meaning
good	0.20996	review	near LOW threshold
good	0.37329	review	ambiguous
good	0.21594	review	near LOW threshold
good	0.59424	defective	false reject

Source files: results/random20_final_test/predictions.csv, metrics.json, classification_report.txt. The four rows above are exact five-decimal CSV values.

09 Speed: claim only what was measured

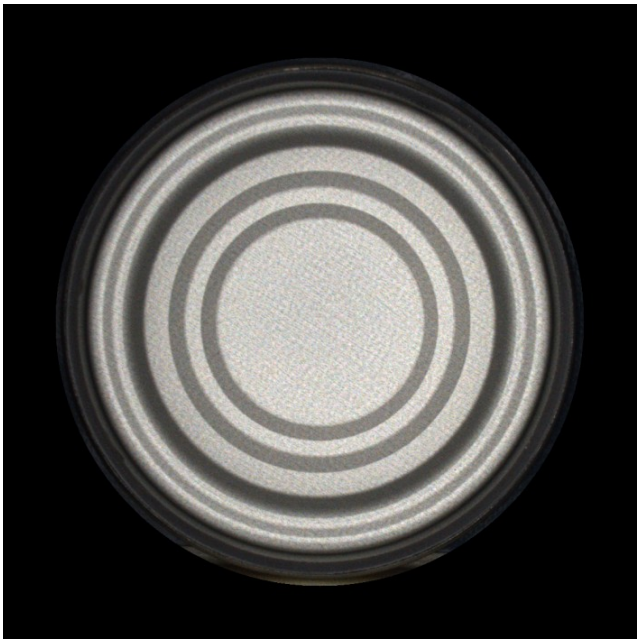
Final script began its timer after loading the model and collecting file paths. It then ran batched model.predict calls, computed decision scores and copied output images to class folders. Timer stopped before writing predictions.csv, metrics.json, the report and confusion matrix. The captured result is 4.185 seconds for 261 pre-masked images (~62.36 images per second). GPU inference alone is faster but is not a plant-level throughput metric.

This test does NOT measure camera acquisition, image trigger timing, lid locating/masking from raw source images, raw image file writing, conveyor/PLC or pneumatic reject latency, nor a sustained live streaming workload. The target is at least 25 end-to-end items/second (about 10.44 seconds for 261). It still needs a separate reproducible raw-input full-pipeline benchmark and factory pilot.

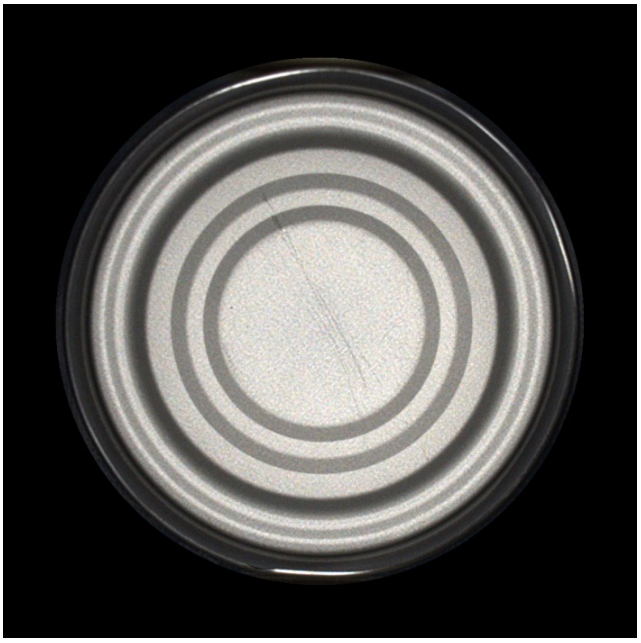
10 Visual examples included in the project archive

The final test ZIP contains sample copies from all three output folders. They are output images without overlaid probability labels; probabilities reside in the CSV. For customer-facing illustrations annotate copies rather than altering evidence files.

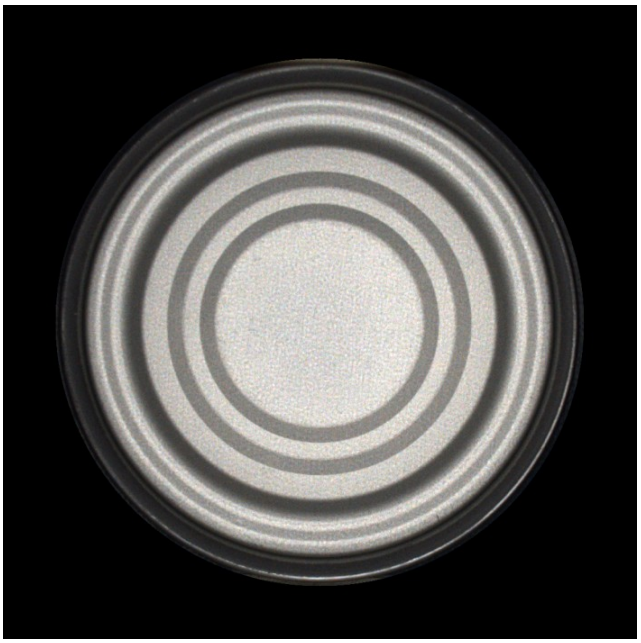
GOOD | Actual good lid accepted as good



DEFECTIVE | Actual defective lid classified defective



REVIEW | Actual good lid routed for operator review



11 Common execution errors and the fixes

Observed issue	Cause / correct handling
Windows torch multiprocessing bootstrap RuntimeError	Workers=4 plus top-level training code caused spawn/import re-entry. Final successful training used workers=0; a future script should also wrap main() in if __name__ == "__main__":.
Attempt to create val: FileNotFoundError	A previous faulty helper deleted from train with Path.unlink() before copying! Recovered from original 80/20 source; corrected helper uses shutil.move. Avoid any destructive operations without manifest/backup.
Training crash when only train folder passed	Ultralytics classification needs parent dataset with train/val. Do not set DATASET to random80/train.
AttributeError: dict has no attribute index	result.names is a mapping, e.g. {0:'defective', 1:'good'}; use {name:idx for idx,name in result.names.items()} to find class indices.
ValueError: Target is multiclass but average='binary'	Mixed true labels and REVIEW actions cannot be directly passed to default binary metrics. Evaluate REVIEW separately; use explicit target labels and appropriate denominators.
UndefinedMetricWarning on macro 3-class report	REVIEW is an abstention action, not a third ground-truth class. Avoid interpreting macro 3-class score as classifier quality.
WARNING 'half' deprecated	Warning during inferencing, not a failed test. Actual script ran with half=True; verify supported future precision API before updating.
Stopping a runaway terminal command	Ctrl+C in Windows PowerShell/VS Code terminal.

12 Existing codebase: file roles

File	Purpose / status
src/00_check_environment.py	CUDA, dependency and environment checks
src/rojin_pipeline.py	Geometric lid ROI, masks, I/O and supporting pipeline
src/rojin_pipeline_backup_strict.py	Earlier strict backup of segmentation pipeline
src/01_build_masked.py	Generate masked images; audit contact sheets/overlays, masks and summary
src/02_train_masked.py	Train final 943/103 classification experiment (current ZIP)
src/03_evaluate_masked.py	Older masked_v1 evaluation / legacy threshold rules
src/03_evaluate_random20.py	Earlier evolving 261-image test, had dict and metrics errors
src/03_final_test_random20.py	Final 261-image batch test / CSV / 3 decision folders / reports
src/04_manual_override.py	Manual override handling for segmentation
src/05_verify_and_approve.py	Audit/approval of segmentation outputs
src/06_export_optional_seg.py	Optional segmentation export (not part of final classifier)
src/07_train_optional_seg.py	Optional segmentation train (not part of final classifier)
src/08_calibrate_validation.py	Legacy validation-only threshold exploration on masked_v1
src/09_create_final_dataset_split.py	seed=42 stratified 80% vs 20% final split
src/10_create_val_from_random80.py	10% validation from 80% train pool
src/10_create_val_split.py	Additional/older val splitter; do not assume active
src/11_verify_dataset_integrity.py	SHA-256 consistency, coverage, leakage audit, read-only
src/12_pack_rojin_project_for_proposal.py	Zip relevant source/results/sample images safely
src/evaluate_demo25.py	Previous demo25 small-set experiments
src/evaluate_internet25.py	Internet external small evaluation, limited evidence
src/test_external.py	Additional external test, uses 0.15 / 0.45 thresholds
src/train_n.py, evaluate_n.py	Early YOLO classification baseline and binary evaluation
Other helper scripts	build_test_sets, inspect_dataset, inspect_best, prepare_external_test, download_and_prepare_internet25, check_dataset_leakage

Safe commands to resume work on Windows

```
cd D:\rojin_lid_ai
.\.venv\Scripts\Activate.ps1
# Re-run final test only when needed (does not retrain):
python .\src\03_final_test_random20.py
# Read-only dataset audit:
python .\src\11_verify_dataset_integrity.py
# Create portable source+evidence archive:
python .\src\12_pack_rojin_project_for_proposal.py
```

Re-running tests into existing output folders may overwrite existing copied images and reports. Preserve / version the current finished outputs before any new experiment. Do not rerun training or rebuild data casually.

13 Earlier models, early experiments, and interpretation

- The earliest experiment used the original 1045/197/65 train/val/test split and baseline yolo11n classification scripts; it is not the final 80/20 model.
- An earlier masked_v1 model trained up to 52/60 epochs, best epoch about 37, with 100% observed validation accuracy on 197 images. That validation result is not an external generalization guarantee.

- Because the final test set was re-randomized over all 1307 images, earlier masked_v1 weights may have seen images now in the new holdout. Do not test masked_v1 weights on random20 and call it an independent held-out test.
- An earlier provisional external Internet sample showed around 37.5% accuracy on 16 labeled images, and a prior batch of 25 internet images was classified defective across the board. This is a warning about domain shift, small-sample labels, and model generalization, not an outcome of the final random80_v2 model on Rojin factory images.

14 Proposal: what was requested and what exists

User asks for an ENGLISH industrial proposal aimed at Rojin Taak Agro Industries, informed by three external inspection vendor PDFs, describing precisely this model as an OFFLINE DEMO / PoC for tomato-paste can lids. The proposal should be visually compelling, evidence-based, and expandable subject to demonstration acceptance.

Primary proposal reference PDFs supplied by user

Reference file	Material to borrow conceptually
ibea visual inspection (1)(4).pdf (14 pages)	2018 ibea GmbH proposal addressed to Rojin; customer specification, lid/can pictures, defect example grids, engineering requirements, acceptance
HiControls - mini pictures Feb.2017 - LOW RES(4).PDF (4 pages)	Brochure imagery and inspection capabilities, labels/caps/body, conveyor examples
Metal Printing Offline Inspection(4).pdf (16 pages)	Technical layout, camera/lighting requirements, defect example panels, acceptance testing concepts

Use their structure/quality expectations, not their hardware specifications, company performance numbers, purchased-machine photographs or product claims as evidence of the user's own system.

Latest produced proposal artifacts

```
/mnt/data/Rojin_AI_Can_Lid_Inspection_Proposal_v1.pdf
/mnt/data/Rojin_AI_Can_Lid_Inspection_Proposal_v1.docx
# Earlier draft files also exist:
/mnt/data/Rojin_AI_Can_Lid_Inspection_Proposal_Draft_v1.pdf
/mnt/data/Rojin_AI_Can_Lid_Inspection_Proposal_Draft_v1.docx
```

The assistant reported a 12-page English v1 draft and editable Word, created after examining project ZIP and reference PDFs. They exist in the available artifact workspace. Treat it as a REVIEW DRAFT rather than automatically customer-ready or contractually committed. Future conversation may need these files reattached.

Current versus proposed scope (must remain explicit)

Already demonstrated	Potential later phase (NOT yet built)
Offline pre-masked lid classification, confidence scores, decisions and reports	On-site camera/image acquisition, lighting, raw ROI segmentation benchmarking
Model weights and local validation/test on external public dataset	Pilot with Rojin-specific lids, real defects and site conditions
Image sorting into GOOD / DEFECTIVE / REVIEW	Can body damage, dents, scratches, contamination
Batch inference on RTX 5060	Label presence, print/alignment and label-quality checks
Report CSV, images, confusion matrix	PLC, conveyor, reject mechanism, operator interface and traceability
Software demo	Extension to other Rojin packaging, formats, lines and products

Information still not resolved / do not invent

- Actual Rojin sample images and label/defect definitions; industrial line throughput and production schedule.
- Camera/resolution/lens, lighting constraints, physical position, trigger, PLC protocol, reject time, conveyor speed.
- Factory-specific acceptance criteria: allowable false-accept risk, test volume, recall, review budget and SLA.
- Who submits the proposal (name/company/logo/contact), and whether it is approval-for-demo, an on-site pilot, or a priced contract.
- Pricing, installation dates, warranty/support commitments and safety/compliance tests: no agreed data yet.

15 Evidence-backed claims vs claims to avoid

Supported / allowed with qualification	Not yet supported
Masking produced 1307/1307 outputs; manual audit caveat applies	Every segmentation is pixel-perfect
Image-level stratified 80/20 and SHA-256 zero exact overlap	No near duplicates, temporal similarity or capture-session leakage
Validation top1 near 99% on 103 validation images	99% accuracy on Rojin plant output
Operational 98.47% on 261 masked public held-out images	98.47% on factory / all tomato paste lids
62.36 pre-masked images/s test-script benchmark	Validated 62.36 cans/s full real-time line or 25 end-to-end items/s
Zero observed false accepts of 147 defective public images	Guaranteed zero defective acceptance in future deployment
Two-class classifier and three-action post-processor	A three-class model trained on REVIEW examples

16 Chronological record of the work and major decisions

01 Early project / baseline. Windows offline Rojin can-lid AI project rooted at D:\rojin_lid_ai; hardware RTX 5060; train and evaluate YOLO image classifier.

02 Original data investigation. TinplateLidDATA2024 identified: 1307 images; 571 good and 736 defective; original train/val/test directories not adequate for final holdout specification.

03 Segmentation and preprocessing. Implemented lid-focused masking, black background, masks/overlays/contact sheets and audit. Full run succeeded automatically on 1307 images; approx. 30 edge masks visually imperfect.

04 First training. Earlier masked_v1 model achieved very high validation accuracy on 197 validation images; cannot transfer claims to a reshuffled holdout.

05 User changes evaluation design. Requested stratified random 80/20 across all 1307 images; test data completely untouched throughout training and tuning.

06 80/20 split creation. Used src/09_create_final_dataset_split.py seed42; generated 1046 train pool and 261 independent test.

07 Validation folder requirement. Ultralytics classification needs train AND val. A flawed helper deleted files causing FileNotFoundError. Data recovered from 80/20 split; corrected val script.

08 Final validation split. Train became 943 (412 good/531 defective); val 103 (45 good/58 defective), test remains 261.

09 Windows worker error. workers=4 at top-level caused torch multiprocessing bootstrapping RuntimeError; use workers=0, ideally main guard.

10 Final training. src/02_train_masked.py trained yolo11n_cls_masked_random80_v2: 61 epochs completed, best epoch 41, 99% validation top1.

11 First proposed test script. Individual-image inference attempted; crashed because result.names is dict, not list (.index is invalid).

12 Second test iteration. Fixed class mapping but metrics failed with binary sklearn precision because REVIEW output created class mismatch; clarified selective vs all-image reporting.

13 Final batched test. src/03_final_test_random20.py with batch32 and half=True created output folders and predictions; 261 images in 4.185s, 62.36 images/sec.

14 Metrics and caution. 110 GOOD, 148 DEFECTIVE, 3 REVIEW; 98.47% all-image decisions; 0 false accepts, 1 false reject; macro 3-label sklearn metrics misleading.

15 Data integrity audit. src/11_verify_dataset_integrity.py checks SHA-256 exact content, counts and parent 80/20 membership; PASS all checks.

16 Legacy vs final thresholds. Old 0.15/0.45 present in older scripts; final 0.20/0.50 in current 03_final_test_random20.py, separate from YOLO weights.

17 Proposal scope clarified. User confirmed DATASET IS PUBLIC/EXTERNAL with lids visually similar to Rojin; demo is for lids only, future body/labels/products conditional.

18 Three reference PDFs. ibea proposal to Rojin, HiControls brochure, Shanghai UPG sheet-inspection proposal supplied for style/content guidance.

19 Portable project archive. src/12_pack_rojin_project_for_proposal.py created rojin_proposal_review_package.zip with source code, final weights, CSV, train graph, sampled images and audit.

20 English proposal v1. Produced 12-page English industrial PoC proposal + editable Word; v1 is draft for review before use.

21 Current user request. Preserve full context and key chat milestones in a portable PDF for a new conversation; this is that document.

17 User statements that affect all future work

The following are condensed or exact short excerpts of user requirements, preserved to prevent losing key intent. They are NOT a verbatim full chat transcript.

Keep holdout separate: train and test data must not mix. [میخوام ترین ... تست جدا باشه]

Output every p_good and p_defect between 0 and 1 for every image. [برای هر عکس بهم بگه میزان good و defective ... بین 0 تا 1]

Save actual images in predicted good, defective, review folders. [خروجی ... پوشه های review,good,defective]

Benchmark processing about 261 images within 11 seconds (>=25 images/s goal). [تمام عکس ها رو حدودا در 11 ثانیه خروجی هاشون رو بده]

Verify 80/20 source membership and all segmented images; internal confidence check. [... عکس ها واقعا متفاوت باشند ... تمام عکس ها ... شامل شدن]

Final proposal in English; ONLY tomato paste lid demo; later label/body/other products after approval. [پروپوزال انگلیسی ... نسخه دمو ... فقط بررسی درب های رب ... لیبل ... بدنه ... سایر محصولات]

Use three vendor PDFs for proposal style and inspect current project ZIP. [طبق اونا ... pdf چند تا پروپوزال یا ...]

External public dataset visually resembles factory lids. [یه دیتاست خارجی و عمومیه اما درب ها خیلی شبیه درب رب روژین هست]

Create a portable PDF for understanding the entire project in a new chat. [... کل چت ها و هر بخش مهمی ذخیره کنی pdf داخل یه]

18 Exact tool / code details to preserve

Final training script core settings

```
DATASET = ROOT / "data" / "processed" / "TinplateLidDATA2024_masked_random80"
model = YOLO("yolo11n-cls.pt")
model.train(data=str(DATASET), epochs=100, imgsz=640, batch=32,
        device=0, workers=0, pretrained=True,
        optimizer="AdamW", lr0=0.001, patience=20, amp=True,
        cache=False, project=str(ROOT/"results"),
        name="yolo11n_cls_masked_random80_v2", exist_ok=False,
        plots=True)
```

Final test inference essentials

```
MODEL_PATH = ROOT / "results" / "yolo11n_cls_masked_random80_v2" / "weights" / "best.pt"
TEST_PATH = ROOT / "data" / "processed" / "TinplateLidDATA2024_masked_random20" / "test"
OUTPUT = ROOT / "results" / "random20_final_test"
batch_size = 32
results = model.predict(paths, device=0, batch=batch_size,
                half=True, verbose=False)
# result.names is a dict (e.g. {0:'defective',1:'good'}).
# Use ids from result.names.items(); do NOT call names.index().
```

Actual final terminal summary

```
ROJIN LID AI FINAL RANDOM20 TEST
Loading model...
Total images: 261
{ "images": 261, "total_time_seconds": 4.185, "fps": 62.36,
  "GOOD_predictions": 110, "DEFECT_predictions": 148,
  "REVIEW_predictions": 3, "accuracy_all": 0.9847,
  "precision_macro": 0.6644, "recall_macro": 0.655,
  "f1_macro": 0.6596, "false_accept": 0, "false_reject": 1 }
```

Actual final dataset audit terminal summary

```
Reference: total=1307
Train/val: total=1046    Test: total=261
coverage_match: True
original_80_20_membership_preserved: True
train_test_content_overlap_count: 0
train_val_content_overlap_count: 0
duplicate_sha256_count: 0
missing_from_new_combined_count: 0
extra_in_new_combined_count: 0
expected_counts_ok: True
passed: True
RESULT: PASS - dataset verified
```

19 Suggested next actions for the receiving conversation

1. Open/read the actual 12-page proposal v1 PDF and DOCX; verify all screenshots, plots, tables, thresholds and any business phrasing before editing.

2. Ask user which audience/decision is sought: initial demo endorsement, factory pilot approval or contract proposal. Do not invent price/time/warranty commitments.
3. If available, request genuine Rojin lid, can, packaging and production-line images; clearly label any conceptual layout as such.
4. Perform raw-image end-to-end benchmark including segmentation, classification, I/O and decision. Report sustainable throughput separately from batched pre-masked throughput.
5. If threshold selection needs justification, use ONLY the current 103-image validation set (new script/settings) or genuinely new validation data; do not optimize thresholds on the 261 held-out test already reported.
6. Audit quality of a sample of ~30 questionable masks and evaluate frame/session near duplicates as a future validity improvement; no need to present this internal work in every customer document.
7. For future models: consider a group/session holdout, prevent edge-erasing augmentations, and collect factory-specific real defect examples before claims of production readiness.
8. Re-evaluate customer-sample false accept and false reject, review/coverage, calibrated probabilities, confidence intervals, and stability under changing production lighting.
9. Keep the current final trained weights and evidence archive unchanged while making new prospective experiments under new versioned result directories.

20 Source-of-truth map: where to find evidence

Claim / topic	Most reliable evidence
Actual model hyperparameters	results/yolo11n_cls_masked_random80_v2/args.yaml
Training curve and 61-epoch finish	results.png, results.csv and Pasted text(7).txt
Operational 261-image confusion	results/random20_final_test/predictions.csv
Reported FPS and legacy metrics	results/random20_final_test/metrics.json
Final decision threshold semantics	src/03_final_test_random20.py
Data split source membership	split_report.json and split_manifest.csv
Coverage / exact-content nonoverlap	results/dataset_integrity_audit/integrity_summary.json and issues.txt
Mask generation outcome	data/processed/TinplateLidDATA2024_masked_v1/summary.json
Images and existing 12-page customer draft	rojin_proposal_review_package.zip; Rojin_AI_Can_Lid_Inspection_Proposal_v1.pdf/.docx
Client’s reference offer/style	Three ibea / HiControls / Shanghai UPG PDFs

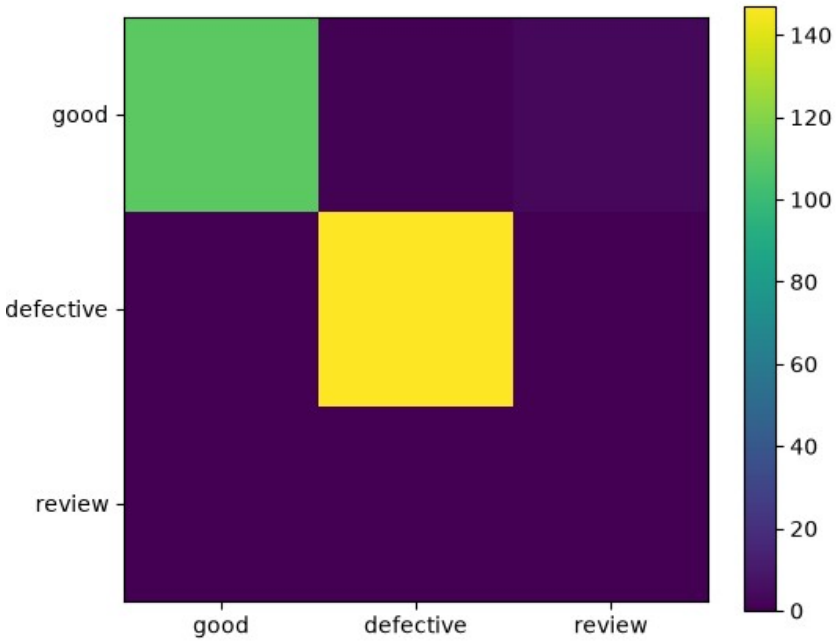
21 Handoff completeness and limitations

This packet was reconstructed from: (a) the accessible conversation history and earlier chat summaries; (b) the original rojin_lid_ai_code_only.zip; (c) the current rojin_proposal_review_package.zip (87 entries); (d) final metrics.json/predictions.csv/args.yaml; (e) dataset SHA-256 audit outputs; and (f) the three user-supplied PDF references and the generated English proposal. It provides unusually detailed continuity but is NOT a full exact transcript of every historic ChatGPT message. Uploaded files and source ZIPs are not embedded wholesale in this PDF.

When a later chat conflicts with this handoff: prefer the original project data files and scripts for factual definitions, actual saved logs for reported measurements, the user’s newest explicit decision for desired goals, and clearly label any newly generated results. Treat the model path and deployment assumptions as local user-machine paths; no assistant in a fresh cloud chat can automatically inspect D:\rojin_lid_ai without a new upload or connected capability.

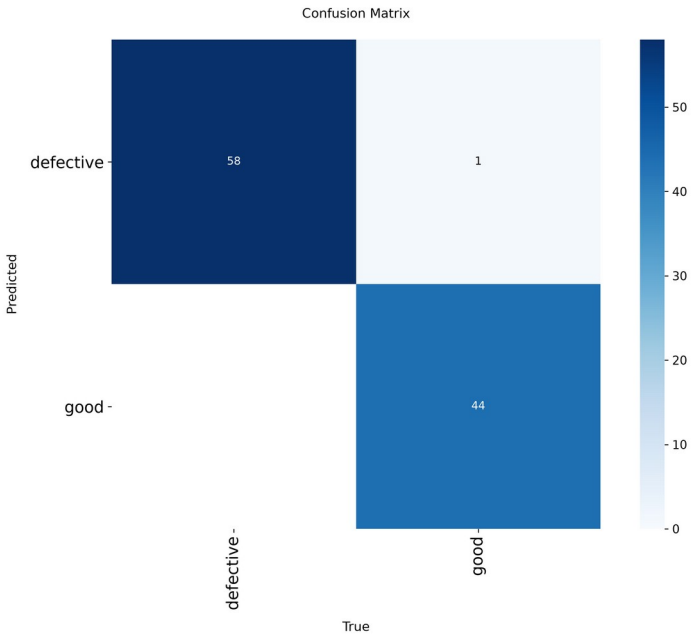
Appendix A Primary visual evidence

Model test confusion matrix (generated by final test script)



Note: the exported matrix is laid out with REVIEW as a predicted category, even though ground truth has only two classes. The verified numeric 2x3 table in Section 08 is the authoritative interpretation.

Additional project-archived training confusion matrix



Figures are taken from the model/project ZIP; they are not evidence of operation at the Rojin plant.

Appendix B Inventory of important attachment names

- rojin_proposal_review_package.zip - portable updated project, code, model weights, train and test reports, sample raw/masked images.
- rojin_lid_ai_code_only.zip - earlier code/results snapshot, useful for tracing legacy evaluation and pipeline variants.
- integrity_summary.json and issues.txt - final independent data integrity audit results.
- Pasted text(7).txt - final full train console log showing epochs and final validation.
- Rojin_AI_Can_Lid_Inspection_Proposal_v1.pdf and .docx - first 12-page English industrial proposal.
- ibea visual inspection (1)(4).pdf - customer-addressed equipment proposal style.
- HiControls - mini pictures Feb.2017 - LOW RES(4).PDF - visual product-brochure reference.
- Metal Printing Offline Inspection(4).pdf - engineering/inspection reference.

END OF HANDOFF DOCUMENT | The next assistant should resume the English proposal using the final model and evidence, not restart from the old baseline.